E-VFU: Multi-Attack Verification for Vertical Federated Unlearning

PETS-style VFU-KD / VFU-GA with extended verification and feature-entanglement discussion

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Undergraduate machine learning project

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Outline

1. Introduction: problem & motivation
2. Dataset and statistics
3. Architecture and training
4. Loss functions and intuition
5. Feature entanglement perspective
6. Related work and SOTA positioning
7. E-VFU verification framework
8. Experiments and results
9. Challenges, conclusions, Q&A

Vertical Federated Learning (VFL)

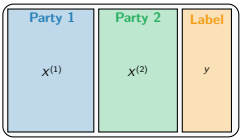
↔ Horizontal FL

Same features, different samples (e.g., many hospitals, same tests, different patients).

↕ Vertical FL

Same samples, different features (e.g., bank + hospital + label holder for the same user IDs).

Parties exchange **embeddings** or gradients — not raw joint feature vectors.



Why unlearning in VFL?

1. **Right to erasure** (GDPR): data must be removed from trained models, not only from databases.
2. Unlearning targets: **client** (party leaves), **feature** (column forgotten), or **sample** (specific rows).
3. Full retraining is costly $\Rightarrow$ approximate methods ($\mathrm{VFU\text{-}KD}$, $\mathrm{VFU\text{-}GA}$, fine-tuning).

Core question

How do we verify that unlearning worked? Weak metrics can declare success while stronger attacks still detect membership or artifacts.

Base paper: Varshney et al. (PETS 2025)

- Feature-level VFL unlearning: **VFU-KD** (distillation) and **VFU-GA** (gradient ascent).
- Active party may retrain a student / run ascent; passive encoders may stay frozen.
- Evaluation includes **output-based** membership inference (MIA).

Why go beyond output MIA?

Logits are not the whole story: gradients and internal representations can still leak membership. Stronger MIAs are a stricter test of whether unlearning actually held.

What we add on top of PETS

1. **Gradient MIA** and **shadow-model** meta-classification alongside output MIA.
2. **Feature entanglement:** correlated columns across parties mean party k's signal may partly live in party j's encoder; we discuss cross-party Jacobian-style diagnostics (see also LASER-VFL on missing features).
3. **Open engineering questions:** secure-aggregation cost, repeated unlearning rounds.

E-VFU: keep PETS-style unlearning; add **four verification attacks** with clear pass/fail-style targets.

Datasets and VFL split (statistics)

Dataset	Samples	Features	Target / balance
UCI Adult	30,162	14	Income >50K: 7,508 vs. 22,654 (∼25% positive)
German Credit	1,000	20	Good vs. bad: 700 / 300
Bank Marketing	41,188	16	Term deposit: imbalanced (minority positive)
Credit Card Default	30,000	23	Default: imbalanced (minority positive)

- Label encoding for categoricals; drop Adult rows with missing values; **per-party standardization**.
- Features partitioned across **2–3 parties**; one active party holds **labels**.
- Stratified train/val/test, seed 42.

Model architecture (Phase 0)

Bottom encoders

Each passive party: MLP on local $X^{(k)} \to$ embedding e_k.

Top model

$[e_1 \| \cdots \| e_K] \to$ MLP $\to$ sigmoid; **BCE** during training.

Unlearning

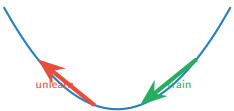
VFU-KD: student top / student bottom + teacher.
VFU-GA: ascent on target subset.

Training (typical): Adam or RAdam, lr $\approx 10^{-3}$, on the order of tens of epochs, batch 256–512.

VFU-GA: intuition

Training: $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$. Unlearning: $\theta \leftarrow \theta + \eta \nabla \mathcal{L}$ on **target** samples.

- Fast; implement by negating the task loss on the forget set.
- Risk: utility drop; may leave **forensic** traces (FUMA-sensitive).



VFU-KD: loss

$$\mathcal{L} \;=\; (1 - \lambda)\, \underbrace{\mathrm{BCE}(p_s, y)}_{\text{fit retain labels}} + \lambda\, \underbrace{\mathrm{KL}(p_s \,\|\, p_t)}_{\text{match teacher}}$$

- Student sees embeddings **without** removed party (or reduced features); teacher is **frozen** pre-unlearning model.
- **Intuition:** BCE keeps task accuracy on retain data; KL preserves decision boundary *as if* the old fusion still had access, while dropping the forgotten party from the forward pass.
- In our code, $\lambda = 0.3$ is a typical distillation weight; other runs use temperature-scaled KL with a different α (same trade-off, different parameterization).

Unlearning methods at a glance

Aspect	VFU-GA	VFU-FT	VFU-KD
Speed	fast	medium	slower
Utility	can drop	good	best (typ.)
Forgetting	strong	medium	strong

Recommendation

Use $\mathrm{VFU\text{-}GA}$ for quick removal when utility loss is OK; use $\mathrm{VFU\text{-}KD}$ when quality matters. **Always** run multi-attack verification afterward.

Feature entanglement: why it matters for VFU

- Shared loss couples all encoders; **correlated** columns across parties let f_j exploit patterns that partly "stand in" for removed party k.
- Removing X_k can cause **collateral** drops for remaining parties (reported $>18\%$ in some missing-feature studies, e.g. LASER-VFL narrative).

Jacobian-based diagnostic (optional extension)

Measure cross-party sensitivity, e.g. $J^{(l)}_{k\to j} = \partial h^{(l)}_j / \partial X_k$, and summarize the spectrum (SVD) to flag high influence *before* unlearning. This complements black-box MIA checks.

Where Jacobian + cross-party + VFL meet

	Jacobian tool	**Cross-party**	**VFL unlearning**
Same-party Jacobian defences (e.g. SensFL)	✓	—	partial
Certified VFU / constraint methods	—	✓	✓
This project	✓	✓	✓

Verification attacks ask whether the model still *remembers*; entanglement analysis asks how tightly parties were coupled before anyone left.

Comparison with selected prior work

Line	**Representative**	**vs. our project**
HFL unlearning	FedEraser, SISA, KNOT	Different partition (rows vs. columns).
VFL unlearning	VFU-LR; Gu et al. few-shot	Other mechanisms; we build on PETS VFU-KD/GA.
PETS 2025	Varshney et al.	Unlearning algorithms we implement.
This project	E-VFU + Jacobian angle	Heavier **verification stack**; optional entanglement discussion.

E-VFU in one slide

Phase 0

Train VFL model; run $\mathrm{VFU\text{-}KD}$ and/or $\mathrm{VFU\text{-}GA}$ as in PETS-style setup.

Phase 1 — four auditors

1. **Output MIA** (Varshney baseline): target AUC ≈ 0.5 on unlearned samples.
2. **Gradient MIA**: norms + layer stats + spectral features $\Rightarrow$ Random Forest.
3. **FUMA**: forensic before/after detector on (prediction, loss, gradient-change) features; target low detectability.
4. **Shadow models**: model-level meta-classifier — does unlearned model look "retain-only"?

What we intended vs. what we observed (rubric-friendly)

Intent (verification goal)

After unlearning, the model should be **useful** on test data but **indistinguishable** from a model that never saw the target data.

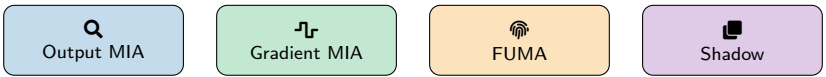
What we implemented

- PETS-style $\mathrm{VFU\text{-}GA}$ and $\mathrm{VFU\text{-}KD}$ (plus FT baseline).
- Multi-attack auditing: Output-MIA, Gradient-MIA, FUMA, Shadow.
- Cross-dataset experiments (Adult, German, Bank, Credit Card).

What we observed

- Utility often stays high, but **verification signals can disagree**.
- Gradient/forensic/model-level attacks can reveal traces even when logits-based MIA is near chance.
- Stronger auditing reduces false "pass" conclusions.

Why four attacks?



Covers *outputs*, *gradients*, *probe behavior*, and *global model fingerprint*. Passing only (1) does not imply passing (2)–(4).

Success criteria (summary)

Component	**Metric**	**Desirable outcome**
Output MIA	AUC	≈ 0.5
Gradient MIA	AUC / $\Delta\|g\|$	near chance + large shift on targets
FUMA	Detection rate	low (e.g. $< 5\%$)
Shadow	P(full data)	< 0.5 for unlearned model

Experimental setup

- 4 public tabular datasets; VFL simulated by splitting feature columns across parties; binary classification.
- **Forget set:** typically 5% of training indices (sample unlearning in Part 4), or a removed party / column group (feature/client unlearning in earlier notebooks).
- **Unlearning methods:** $\mathrm{VFU\text{-}GA}$ (ascent on forget loss), $\mathrm{VFU\text{-}KD}$ (retain-only student with teacher guidance), $\mathrm{VFU\text{-}FT}$ (retain-only fine-tune baseline).
- **Verification:** run Output-MIA, Gradient-MIA, FUMA, Shadow after each method; compare trends across datasets.

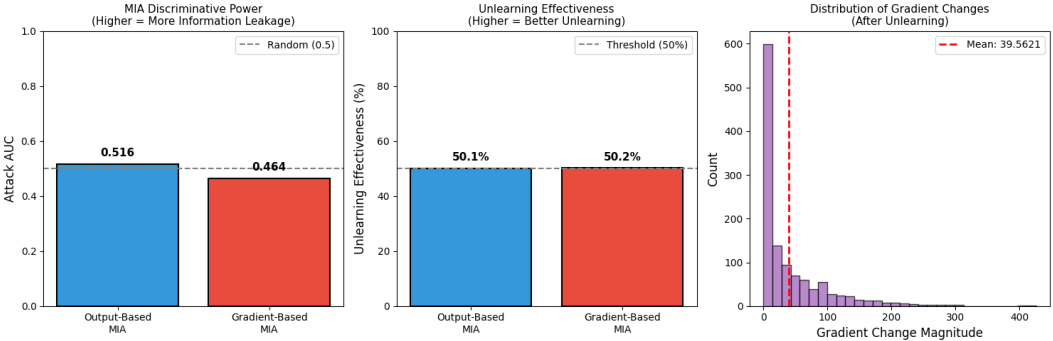
Training history (before unlearning)

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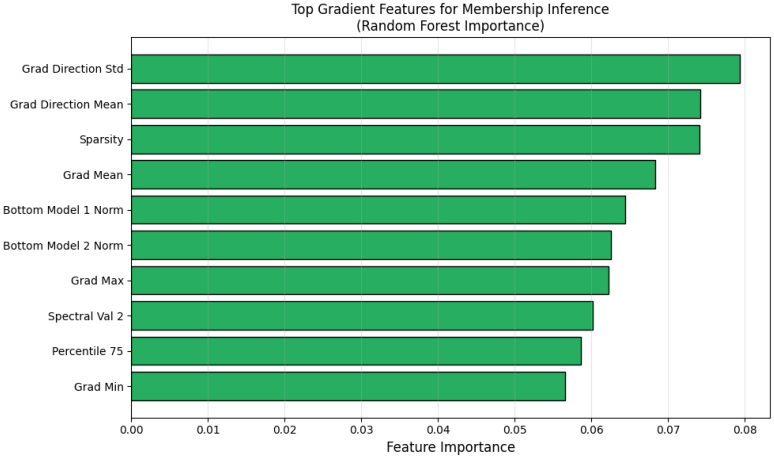
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Training loss and validation accuracy / F1 / AUC on UCI Adult before unlearning.

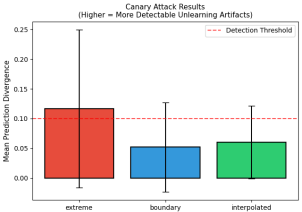
Gradient vs. output MIA

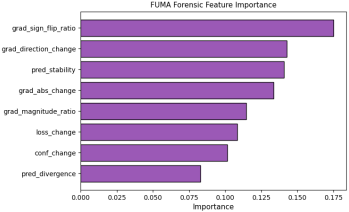


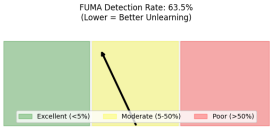
Gradient features (importance)

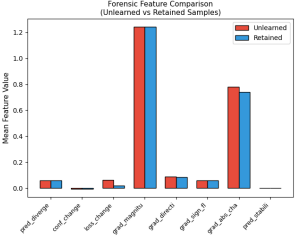


FUMA (forensic)

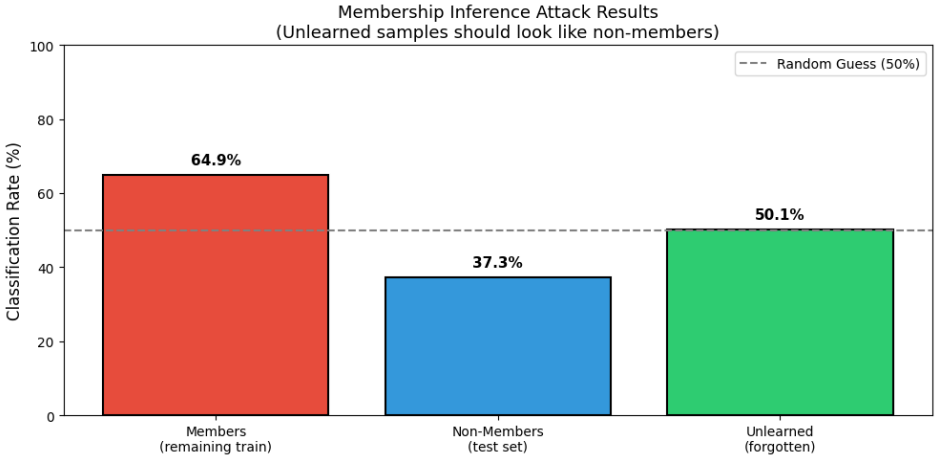




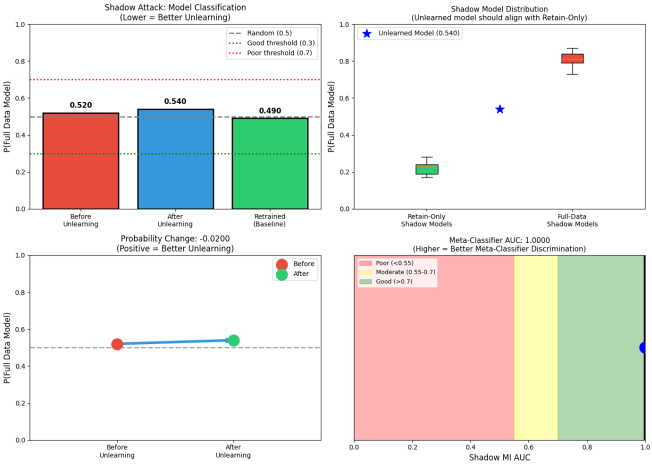




Output-based MIA



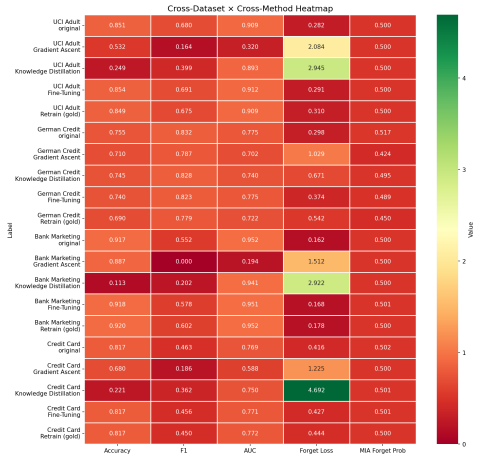
Shadow model attack



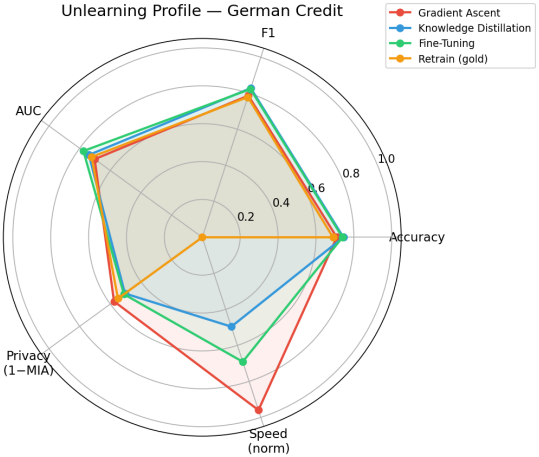
Part 4: Cross-dataset summary (utility + verification)

What this adds beyond Phase 1 on one dataset

- Same unlearning methods, but stress-tested on **four** datasets.
- Lets us see which verifier is **stable** vs. dataset-sensitive.
- Supports a supervisor-friendly conclusion: "verification must be multi-attack and multi-dataset."



Part 4: Method trade-offs (radar view)



Radar plots summarize the utility–verification trade-off per method (higher utility, lower membership/detectability is better).

Comprehensive verification

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Phase 1 dashboard

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Challenges and discussion

- **Threat model mismatch:** auditors must match attacker capabilities (gradient access, shadow training budget).
- **Utility–privacy:** VFU-GA is fast but brittle; VFU-KD needs teacher / history and compute.
- **Entanglement:** correlated cross-party features complicate "complete" removal; Jacobian diagnostics help prioritize risk.
- **Deployment:** SMPC, async parties, repeated unlearning rounds largely **open** for future work.

Summary

- Built a VFL training and unlearning pipeline consistent with Varshney et al. (PETS): VFU-KD and VFU-GA.
- Added a **multi-attack verification framework** (Output-MIA, Gradient-MIA, FUMA forensic detector, Shadow meta-classifier) to reduce false "pass" conclusions.
- Extended evaluation to multiple public datasets (Part 4) and added cross-dataset summary figures.
- Discussed feature entanglement (cross-party Jacobian viewpoint) as a complement to black-box audits.

References

- Varshney et al., PETS 2025 (VFU-KD, VFU-GA).
- Geiping et al., NeurIPS 2020 (gradient leakage in federated learning).
- Shokri et al., IEEE S&P 2017 (membership inference; shadow models).
- FedEraser; SISA (horizontal federated unlearning, context).
- LASER-VFL; certified VFL unlearning (missing features and guarantees).

Thank you

Questions?